

RF and Microwave Engineering Numerical Companion

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · Chapter 7, every radiation-zone calculation, 2069–2082 BS

7. Microwave Antennas and Propagation: Numerical Problems

2 solved problems covering **every radiation-zone calculation in 24 papers** · 81 Bh, 82 Ba, and the microwave-oven zones of 74 Ma and 77 Ch

READ THIS FIRST

Chapter 7 has one calculation: where the three radiation zones of an antenna begin and end. It is always a 3 or 4 mark part of a longer theory question.

- **The procedure:** (1) $\lambda = c/f$, (2) check $D > \lambda$, (3) $R_1 = 0.62\sqrt{D^3/\lambda}$, (4) $R_2 = 2D^2/\lambda$, (5) state the three zones and which are hazardous. Problem 1 shows every line.
- **Do step 2.** When the antenna is smaller than a wavelength (82 Ba's quarter-wave antennas) the formulas put the "far field" inside the reactive field. Give the formula result, then the small-antenna boundaries $\lambda/2\pi$ and 2λ (Problem 2).
- **Units:** keep λ and D in metres; the answers come out in metres.
- Every number below is asserted in `src/ant.py`, which also cross-checks the 81 Bh zone answer against an ICNIRP compliance distance.

Problem 1: Hazardous and safe zones of an antenna larger than a wavelength (the full method) 4 3

4

81 Bh solved in full · 74 Ma, 77 Ch (microwave oven) in the sibling table

QUESTION AS PRINTED 81 Bh [4+4]

Approximate the hazardous and safe zone for BTS tower operating at 2600MHz with antenna size of 0.5 m. Explain, how non-radiating radiation is hazardous to the human body?

The second part is theory: §7.2, "how is non-radiating radiation hazardous".

Given

- $f = 2600 \text{ MHz} = 2.6 \times 10^9 \text{ Hz}$, antenna largest dimension $D = 0.5 \text{ m}$, $c = 3 \times 10^8 \text{ m/s}$.

Formulas

$$\lambda = \frac{c}{f}, \quad R_1 = 0.62\sqrt{\frac{D^3}{\lambda}}, \quad R_2 = \frac{2D^2}{\lambda}$$

$R < R_1$ reactive near field, $R_1 < R < R_2$ radiating near field (Fresnel), $R > R_2$ far field. Valid when $D > \lambda$.

Step-by-step

#	Quantity	Substitution	Value
1	wavelength λ	$\frac{3 \times 10^8}{2.6 \times 10^9}$	0.1154 m
2	validity check	$D = 0.5 \text{ m}$ vs $\lambda = 0.1154 \text{ m}$	$D = 4.3\lambda > \lambda \checkmark$
3	D^3/λ	$\frac{0.5^3}{0.1154} = \frac{0.125}{0.1154}$	1.0833 m ²
4	R_1	$0.62\sqrt{1.0833} = 0.62 \times 1.0408$	0.645 m
5	R_2	$\frac{2 \times 0.5^2}{0.1154} = \frac{0.5}{0.1154}$	4.33 m

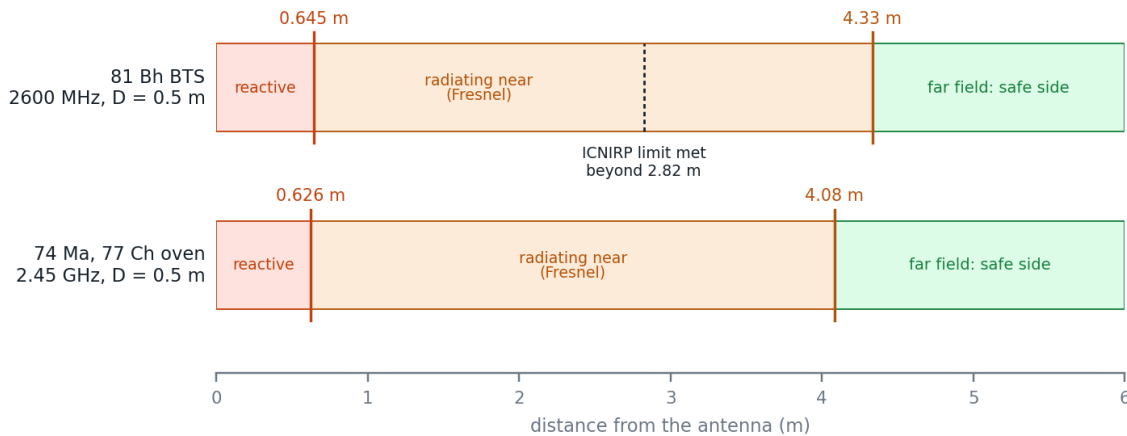
Why step 2: the formulas describe an aperture larger than a wavelength. If $D < \lambda$ they must not be used (Problem 2).

Final answer

Zone	Distance from antenna	Verdict
Reactive near field	$R < 0.645$ m	hazardous (stored-energy coupling)
Radiating near field	$0.645 < R < 4.33$ m	hazardous (full beam power, not yet spreading)
Far field	$R > 4.33$ m	safe side: power density falls as $1/R^2$

Hazardous zone: within 4.33 m of the antenna Safe zone: beyond 4.33 m

The zone figure, to scale



What “approximate the zones” and the oven’s “illustrate the radiation fields” want drawn: the three regions along the distance from the source, boundaries marked. The dashed line is the ICNIRP compliance distance from the cross-check below. Generated by `zonefig.py`, which checks every boundary against this page.

Cross-check against the exposure limit verified/added

- The paper gives no power, so assume a typical sector: $P_t = 20$ W into a $G = 17$ dBi = 50.1 panel $\rightarrow$ EIRP = $20 \times 50.1 = 1002$ W.
- ICNIRP 1998 public limit above 2 GHz: $S_{lim} = 10$ W/m² (§7.4).
- Compliance distance $R = \sqrt{\frac{\text{EIRP}}{4\pi S_{lim}}} = \sqrt{\frac{1002}{4\pi \times 10}} = \sqrt{7.98} = 2.82$ m (occupational limit 50 W/m²: 1.26 m).
- At the far-field boundary $S = \frac{1002}{4\pi \times 4.33^2} = 4.25$ W/m², under half the limit.
- So the zone answer (4.33 m) is the **more conservative** of the two: anyone kept beyond the far-field boundary is also within the limit. Every value: `ant.py`.

Sibling instances: same method, only the numbers change

Paper	Given	λ	Reactive / Fresnel / far	OK?
81 Bh (above)	BTS, 2600 MHz, $D = 0.5$ m	0.1154 m	< 0.645 m / 0.645 to 4.33 m / > 4.33 m	yes
74 Ma, 77 Ch	microwave oven, 2.45 GHz, $D = 0.5$ m (assumed, as the deck does)	0.1224 m	< 0.626 m / 0.626 to 4.08 m / > 4.08 m	yes

QUESTION AS PRINTED 74 Ma [4+4]

What are the different types of electromagnetic radiation hazard? Illustrate radiation fields of a microwave oven.

QUESTION AS PRINTED 77 Ch [3+3]

How are microwaves hazardous to humans? Define different radiation zones of a microwave oven.

- Oven working, step by step: $\lambda = 3 \times 10^8 / 2.45 \times 10^9 = 0.1224$ m; $D^3 / \lambda = 0.125 / 0.1224 = 1.021$; $R_1 = 0.62 \times 1.010 = 0.626$ m; $R_2 = 0.5 / 0.1224 = 4.08$ m. The deck prints 0.6266 m and 4.085 m from the same numbers.
- Add the oven-specific point: the metal cavity and door mesh keep leakage below 1 mW/cm² at 5 cm, so the zones apply to **leakage** only (§7.1). Draw the zone figure with the oven at the source end.

Problem 2: Radiation fields of a small (quarter-wave) antenna, 900 MHz vs 1800 MHz 4

82 Ba · solved separately because the Problem-1 formulas break down for $D < \lambda$

QUESTION AS PRINTED 82 Ba [4+2+2]

Compare radiation fields around a mobile BTS operating at 900 MHz and 1800 MHz frequency bands with their quarter-wavelength antennas. Explain how microwave EMR becomes hazardous to humans; and mention its IEEE safety standard.

Parts two and three are theory: §7.2 and §7.4 (IEEE C95.1).

Given

- $f_1 = 900$ MHz, $f_2 = 1800$ MHz, antenna length $D = \lambda/4$ at each frequency.

Method

- First apply the textbook formulas, as the examiner expects. With $D = \lambda/4$ they reduce to

$$R_1 = 0.62 \sqrt{\frac{(\lambda/4)^3}{\lambda}} = \frac{0.62 \lambda}{8} = 0.0775 \lambda, \quad R_2 = \frac{2(\lambda/4)^2}{\lambda} = \frac{\lambda}{8} = 0.125 \lambda$$

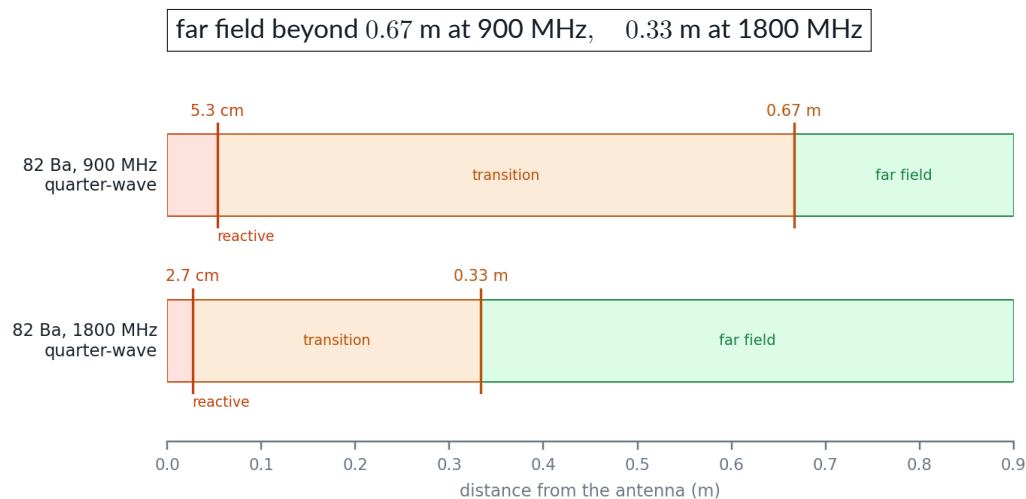
- Then check validity: $D < \lambda$, and $R_2 = 0.125 \lambda$ is inside $\lambda/2\pi = 0.159 \lambda$, where the reactive field of any small antenna still dominates. So the “far field” of the formula is not a far field. Use the small-antenna boundaries verified/added : reactive out to $\lambda/2\pi$, far field beyond about 2λ , a transition region between.

Step-by-step

Quantity	Formula	900 MHz	1800 MHz
wavelength	$\lambda = c/f$	0.3333 m	0.1667 m
antenna length	$D = \lambda/4$	8.33 cm	4.17 cm
textbook reactive boundary	$R_1 = 0.0775 \lambda$	2.58 cm	1.29 cm
textbook far-field boundary	$R_2 = 0.125 \lambda$	4.17 cm	2.08 cm
validity	$D > \lambda$?	no	no
reactive near field (use)	$R < \lambda/2\pi$	$R < 5.3$ cm	$R < 2.7$ cm
transition (radiating near)	$\lambda/2\pi$ to 2λ	5.3 to 66.7 cm	2.7 to 33.3 cm
far field (use)	$R > 2\lambda$	$R > 0.67$ m	$R > 0.33$ m
public limit, FCC	$f/1500$ or 1.0 mW/cm ²	0.6 mW/cm ²	1.0 mW/cm ²
public limit, ICNIRP	$f/200$ W/m ²	4.5 W/m ²	9 W/m ²

Final answer: the comparison

- Every zone boundary scales with λ , so at 1800 MHz each zone is **half** the size of the 900 MHz one:



The comparison drawn to one scale: every 1800 MHz boundary is half the 900 MHz one. From `zonefig.py`.

- Both antennas are electrically small, so the near-field region is tiny: anyone more than a metre away is in the far field of either, and exposure there is set by power and distance ($S = P_t G / 4\pi R^2$), not by zone.
- **900 MHz**: longer wavelength penetrates deeper into tissue (several cm) and is nearer the body-resonance region → **tighter limit** (0.6 mW/cm²).

- **1800 MHz:** absorbed more superficially → limit relaxes to 1.0 mW/cm^2 . Higher path loss → smaller cells, more sites, lower power per site.

The same break-down hits the deck's second example (150 MHz, $D = 0.5 \text{ m}$, $\lambda = 2 \text{ m}$): it prints reactive $< 0.155 \text{ m}$ and far field $> 0.25 \text{ m}$, but 0.25 m is inside $\lambda/2\pi = 0.32 \text{ m}$. The small-antenna rule gives reactive $< 0.32 \text{ m}$, far field $> 4 \text{ m}$. The formulas cross over at $D = 0.28\lambda$ ([ant.py](#)).

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.